

6 (a) total resistance in series = $2R$ M1
total resistance in parallel = $\frac{1}{2}R$ A0 [1]
ratio is $2R / \frac{1}{2}R = 4$ (allow mark if clear numbers in the ratio)

(b) at 1.5 V, current is 0.10 AC1
resistance = $V/I = \frac{1.5}{0.1}$
= 15Ω A1 [2]
(use of tangent or any other current scores no marks)

(c)

	p.d. across each lamp / V	resistance of each lamp / Ω	combined resistance / Ω
series	1.5	15	30
parallel	3.0	20	10

column 1A1
columns 2 and 3: max 3 marks with -1 mark for each error or omissionA3 [4]

(d) (i) ratio is 3(allow e.c.f.)A1 [1]

(ii) resistance increases as potential difference increasesB1
increasing p.d. increases currentB1
current increases non-linearly so resistance increasesB1 [3]

[Total: 11]

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7 (a) either forms of some element